

## Code-Layout, Readability and Reusability

|                      |              |
|----------------------|--------------|
| <b>Date</b>          | 12 June 2024 |
| <b>Team ID</b>       | XXXXXX       |
| <b>Project Name</b>  | Rising Water |
| <b>Maximum Marks</b> | 5 Marks      |

### Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

### Code Layout Checklist:

| S.No | Code Quality Parameter    | Description                                   | Followed (Yes / No / Partial) | Remarks |
|------|---------------------------|-----------------------------------------------|-------------------------------|---------|
| 1    | Consistent Indentation    | Uniform spacing/tabs used throughout the code | Yes                           |         |
| 2    | Proper File Structure     | Files and folders are logically organized     | Yes                           |         |
| 3    | Meaningful Variable Names | Variables reflect their purpose clearly       | Yes                           |         |
| 4    | Function / Method Names   | Functions are descriptively named             | Yes                           |         |
| 5    | Code Comments             | Inline and block comments explain logic       | Yes                           |         |
| 6    | Modular Design            | Code is split into reusable functions/modules | Yes                           |         |
| 7    | No Redundant Code         | Duplicate or unused code is removed           | Yes                           |         |
| 8    | Error Handling            | Exceptions and errors are handled gracefully  | Yes                           |         |

### Reusable Components / Modules:

| S.No | Component / Module Name | Language / Technology | Where Reused      | Reusability Level (High / Medium / Low) |
|------|-------------------------|-----------------------|-------------------|-----------------------------------------|
| 1    | Prediction Function     | Python                | Flask & Streamlit | High                                    |

|   |                     |              |                 |        |
|---|---------------------|--------------|-----------------|--------|
| 2 | Scaler              | Scikit-learn | Prediction      | High   |
| 3 | Random Forest Model | Scikit-learn | Prediction      | High   |
| 4 | HTML Template       | HTML/CSS     | Web Pages       | Medium |
| 5 | Input Validation    | Flask        | Prediction Form | Medium |

Overall Code Quality Assessment:

| Aspect                   | Rating (1-5) | Comments |
|--------------------------|--------------|----------|
| Code Layout & Structure  | 5            |          |
| Readability              | 5            |          |
| Reusability              | 5            |          |
| Documentation / Comments | 4            |          |
| Overall Score            | 5            |          |